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$$\begin{aligned}\lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{e^x - e^{-x} - 2x} &= \frac{0}{0} \\&\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\cos x + x(-\sin x) - \cos x}{e^x + e^{-x} - 2} \\&= \lim_{x \rightarrow 0} \frac{-x \sin x}{e^x + e^{-x} - 2} = \frac{0}{0} \\&\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-\sin x - x \cos x}{e^x - e^{-x}} \\&= \lim_{x \rightarrow 0} \frac{-(\sin x + x \cos x)}{e^x - e^{-x}} = \frac{0}{0} \\&\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-(\cos x + \cos x + x(-\sin x))}{e^x + e^{-x}} \\&= \lim_{x \rightarrow 0} \frac{-(2 \cos x - x \sin x)}{e^x + e^{-x}} \\&= \frac{-(2 \cdot 1 - 0 \cdot 0)}{1 + 1} = \frac{-2}{2} = -1\end{aligned}$$

References